BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LARKANA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2017

PHYSICS

Past Paper 2017

Retyped faithfully from the original board paper — same questions, no changes.

Part – A MULTIPLE CHOICE QUESTIONS (MCQs Type A)

Time Allowed: 20 Minutes | **Maximum Marks:** 16

Note: Identify the correct answer of each (MCQ) from its given four options and write it on the answer script/copy.

Q1. Energy stored per unit volume inside the solenoid is called.....

(a) Charge density
(b) Volume Charge density
(c) Energy density
(d) None

Q2. If the motor is over loaded then the magnitude of "Back emf ".....

(a) Increases
(b) Decreases
(c) Remains same
(d) None

Q3. Farad is not equivalent to.....

(a) Cv^2
(b) J/V^2
(c) Both
(d) None

Q4. The reciprocal of conductivity is called.....

(a) Resistance
(b) Resistivity
(c) Power dissipation
(d) Electric current

Q5. An electric heater of 1000 watt operates for 5 minutes, the heat produced is.....

(a) $5x10^3J$
(b) $3x10^5J$
(c) 200 J
(d) None

Q6. Electric lines of force about a negative point charge are.....

(a) Circular, clock wise
(b) Radial inwards
(c) Both
(d) None

Q7. A device works on the basis of torque of a couple is.....

(a) Generator (b) Motor

(c) Transformer (d) None

Q8. A metal sheet with a circular hole is heated, The hole will.....

(a) Contract (b) Expand

(c) Remain un affected (d) None

Q9. If the radius of first orbit of hydrogen atom is 0.53Å. The radius of second orbit will be .....Å.

(a) 2.12 (b) 0.212

(c) 21.2 (d) None

Q10. The permissible energy levels taken in groups are called....

(a) Permissible level energy (b) Energy band

(c) Conduction band (d) None

Q11. The substances with resistivity of the order of 10^{-8} oh, meter are called.

(a) Insulators (b) Semi conductors

(c) Conductors (d) None

Q12. The production of x-rays is a reverse process of.....

(a) Pair production (b) Compton effect

(c) Photo electric effect (d) All

Q13. Which is not magnetic material.....

(a) Iron (b) Nickel

(c) Silver (d) None

Q14. .....are two halves of a ring that acts as commutator.

(a) Slip rings (b) Split rings

(c) Both (d) None

Q15. The hypothesis regarding dual nature of particles was proposed by.....

(a) Germer (b) de Broglie

(c) Heisen berg (d) None

Q16. The Nuclear radiation with highest velocity is.....

(a) Alpha (b) Beta

(c) Neutron (d) None

Part – B:- Constructed Response Questions (CRQs) / (Short Answer Questions)

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 45

> Note: Attempt any NINE of the following questions.

Q1. How can radiation from radioactive substance harm us?

Q2. Why does laser usually emit only one particular colour rather than several colours?

Q3. Can a charged particle at rest be set in the motion with a power full magnet?

Q4. Why is a volt meter connected parallel to the points between which voltage to be measured?

Q5. Why is potentiometer is considered one of the most accurate voltage measuring device?

Q6. How could you show that the electro static force on charged object is independent of its mass?

Q7. In the photoelectric effect the energy of a photoelectron is always less than that of energy of incident photon why?

Q8. Why Ohms' law is not applicable to the Tungston filament of an electric bulb?

Q9. How Breeder factor is different from other nuclear reactors?

Q10. Is it possible to create a single electron from energy? Explain.

Q11. Why does the specific heat at constant pressure is always greater than the specific heat of gases at constant volume?

Part – C:- (Extended Response Questions (ERQs) / (Long Answer Questions)

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 24

> Note: Attempt any TWO of the following questions.

Q12.

(a) State and explain the first law of thermodynamics. Discuss Isobaric, Isochoric, Isothermal and Adiabatic process on the basis of this law.

(b) The potential difference between the terminals of a battery in open circuit is 2.2 v. When it is connected a cross a resistance of 5.0 ohm the potential falls to 1.8 v calculate the current and the internal resistance of the battery?

Q13.

(a) What is Resistor? How Equivalent resistance is to be find when number of resistors are connected in parallel and in series.

(b) A electron is a accelerated through a potential difference of 50 v calculate its De- Broglie wave length.

Q14.

(a) What is atomic spectrum? How Bohrs' theory of Hydrogen atom explain different spectra of hydrogen atom.

(b) A copper wire of length 2 m and area of cross-section 1.7×10^{-6} m^2 has a resistance of 2×10^{-2} ohm. Find the resistivity.